

DEVOPS





WORKSHOP GOALS AND STRUCTURE

Four-Hour Sessions

Key DevOps Principles for Leaders

Blend of Practical Insights and Demos



PRESENTER INFORMATION

Russell A. Miller, BSME, OOTE, PST, LSS
Green Belt, CCBA

Consultant - Product Development Leader -
Agile Coach - Scrum.org Professional
Trainer - Mechanical & Aerospace Design
Engineer - Analyst

[linkedin.com/in/russellamiller](https://www.linkedin.com/in/russellamiller)



WHAT TO EXPECT FROM THIS WORKSHOP

- Flexibility
- Conversations
- Literacy and awareness on the many principles, tools and practices associated with this thing called “DevOps”
- A priority of focus on human behavior first, technology and tools second
- A lot of talk about organizational culture
- An effort to focus on your own situations and challenges so you can act on what you learn



WHAT NOT TO EXPECT FROM THIS WORKSHOP

- Prescriptions and formulas, rigid processes, step-by-step instructions
- Big overnight transformations
- Perfect solutions that work for everyone
- Extended technical discussions or deep focus on any specific engineering tool



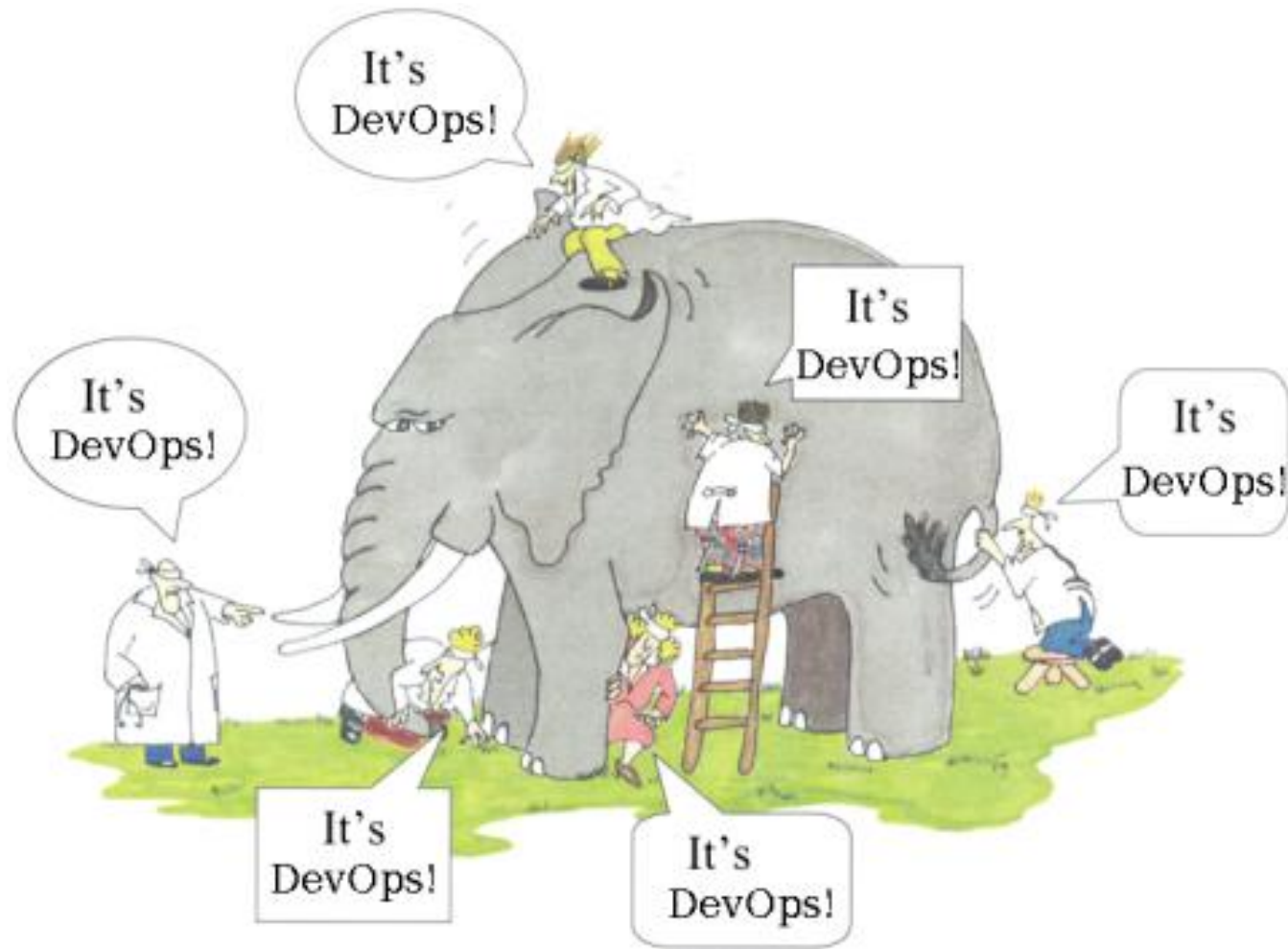
1: INTRODUCTION AND CONTEXT

DevOps Overview

- **DevOps Overview:** High-level view
- **Modern Software Delivery Challenges:** Staying competitive in dynamic markets
- **Aligning Technology with Business Outcomes:** DevOps as a business enabler



WHAT IS DEVOPS REALLY?



WHY DEVOPS MATTERS

Faster Delivery of Features

Reduced Risk and Downtime

Competitive Differentiation



KEY TENETS OF DEVOPS CULTURE

Collaboration Across Silos

Continuous Learning and Experimentation

Ownership and Accountability



2: THE THREE WAYS OF DEVOPS

Flow * Feedback * Learning

- **Streamlining Work from Idea to Production**
- **Gathering Feedback Early and Often**
- **Embracing Blameless Improvement Cycles**



**GENE KIM'S
THREE WAYS OF DEVOPS**

THE FIRST WAY: FLOW

Principles of Flow

- **Make Work Visible**
- **Limit Work in Process (WIP)**
- **Reduce Batch Sizes**
- **Reduce the Number of Handoffs**
- **Continually Identify and Elevate Constraints**
- **Eliminate Waste in the Value Stream**



MAKE WORK VISIBLE

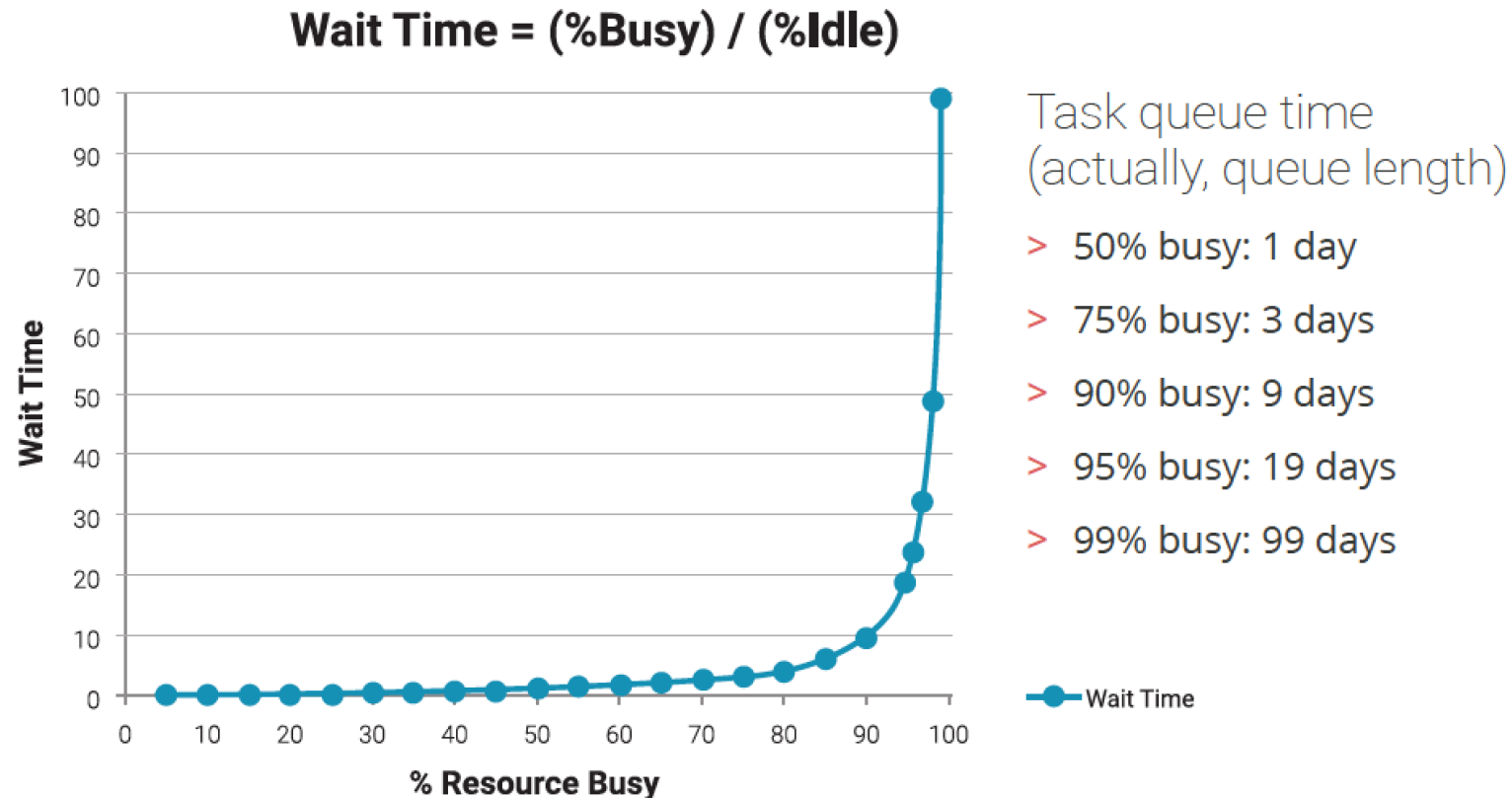
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Make work visible so we all know we are on the same page.

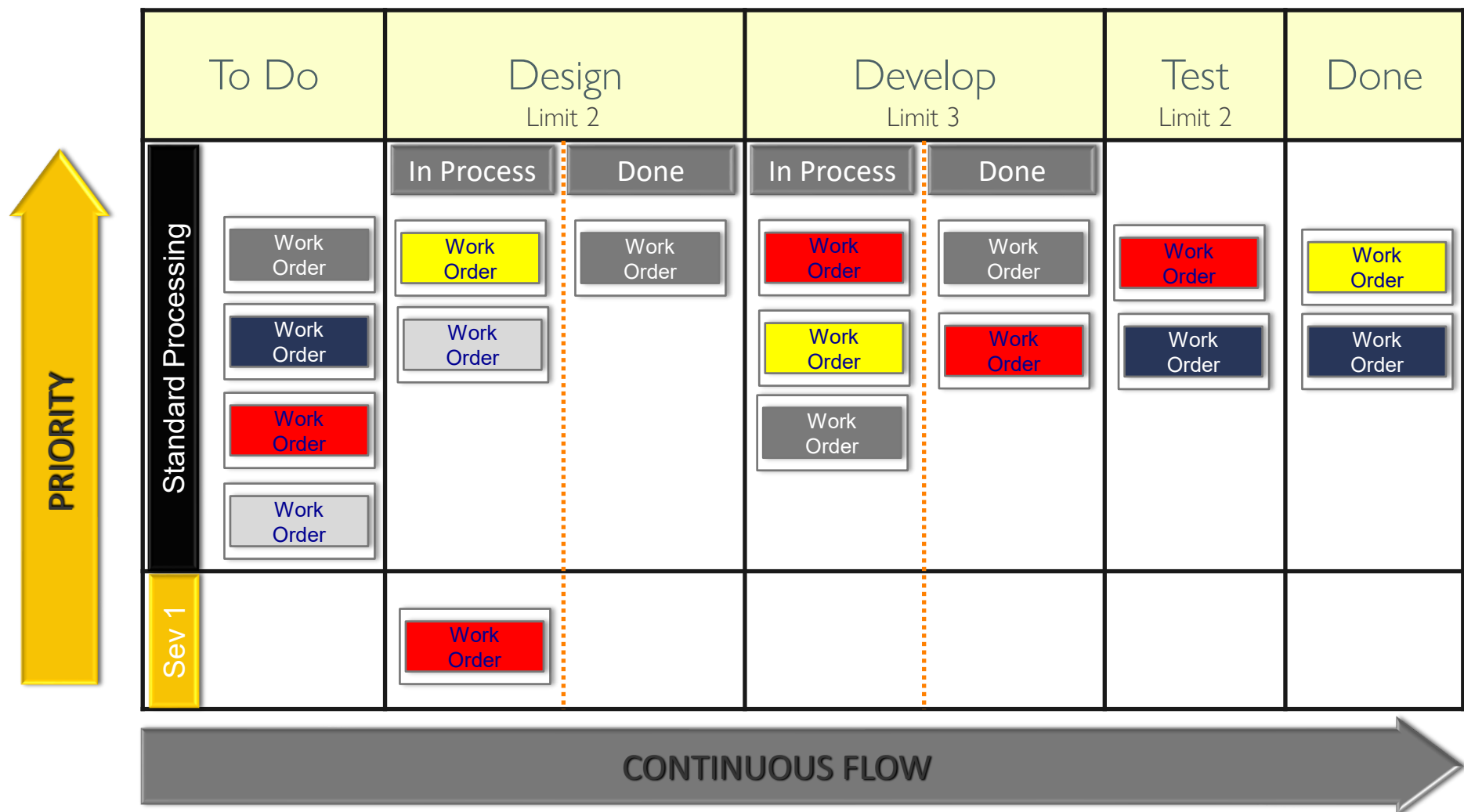


LIMIT WORK IN PROGRESS (WIP)

Don't waste hours managing a 15 minute task



KANBAN WITH WIP LIMITS



REDUCE BATCH SIZES

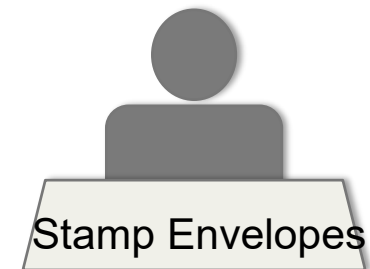
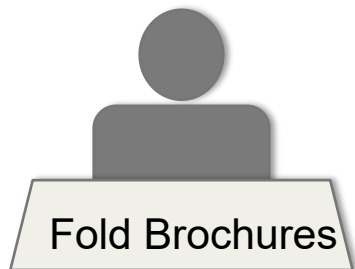
Benefits:

Lower WIP

Less variability in Flow

Reduced Lead Time (Reduced Waiting)

Better Quality (Reduced Rework)



Visualize: 1 batch of 100 - 10 batches of 10 - 100 singles

REDUCE THE NUMBER OF HAND-OFFS

Handoffs

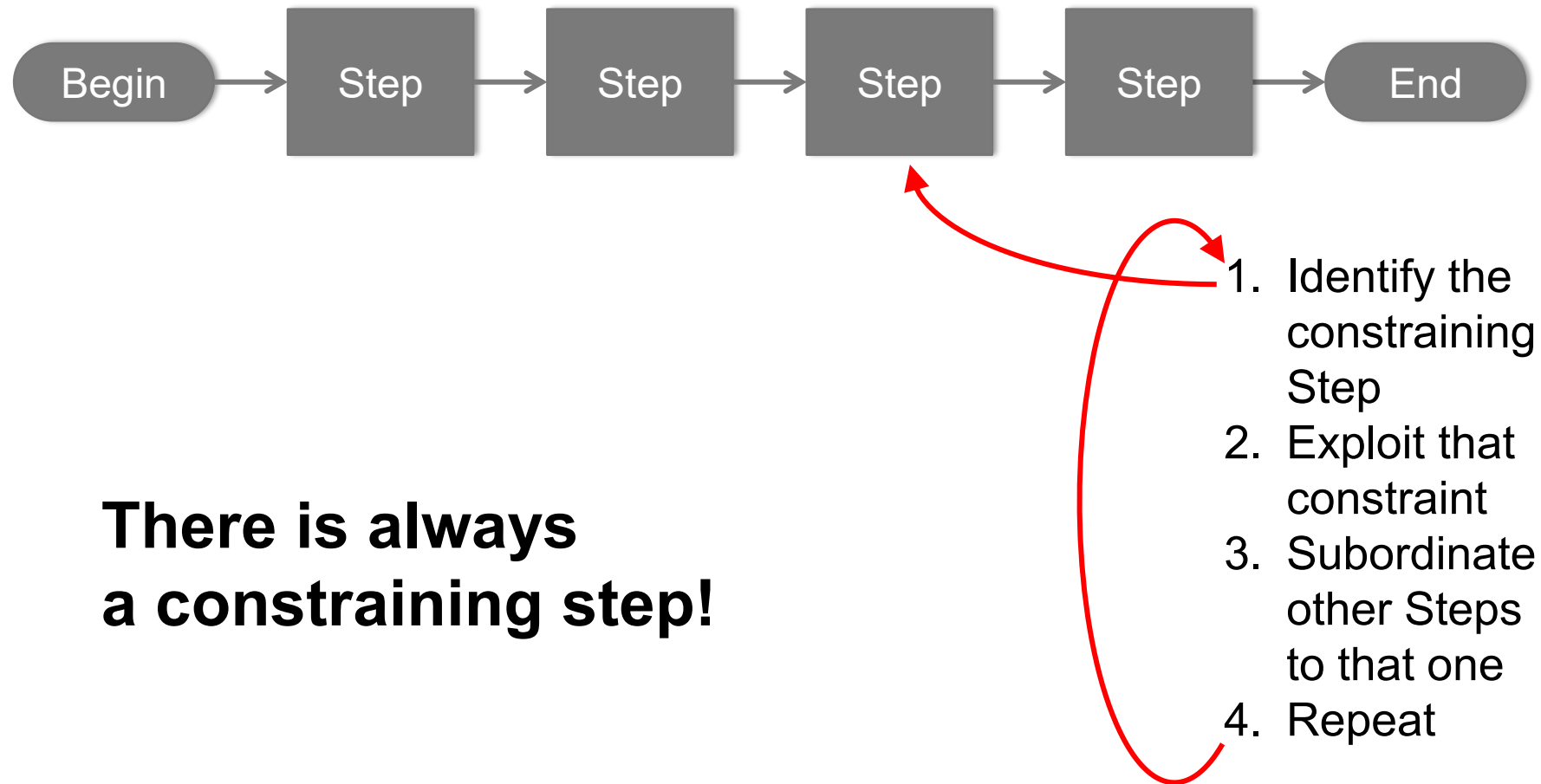
- Increase Lead Time (waiting)
- Reduce %C/A (mistakes)

Reduce Handoffs:

- Cross-functional Teams
- Automation



CONTINUALLY IDENTIFY & ELEVATE CONSTRAINTS



ELIMINATE WASTE IN THE VALUE STREAM

Toyota's Muda, Muri, Mura & the 8 Wasters



Defects

Efforts caused by rework, scrap, and incorrect information.



Overproduction

Production that is more than needed or before it is needed.



Waiting

Wasted time waiting for the next step in a process.



Non-Utilized Talent

Underutilizing people's talents, skills, & knowledge.



Transportation

Unnecessary movements of products & materials.



Inventory

Excess products and materials being processed.



Motion

Unnecessary movements by people (e.g. walking).



Extra-Processing

More work or higher quality than is required by the customer.

THE SECOND WAY: FEEDBACK

Working Safely Within Complex Systems

Principles of Feedback:

- See Problems As They Occur
- Swarm and Resolve Problems to Build New Knowledge
- Keep Pushing Quality Closer to the Source
- Optimize for Downstream Work Centers



SEE PROBLEMS AS THEY OCCUR



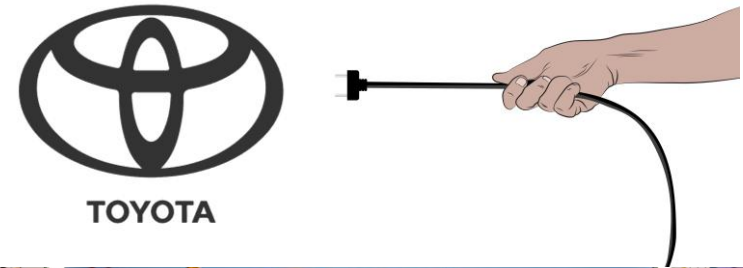
Getting **Feedback Early** and **Often** helps provide **Data** to make **Informed Decisions** to deliver more **Business Value** to the **Customer**

SWARM TO SOLVE PROBLEMS & BUILD KNOWLEDGE

Benefits of Swarming:

- Fix the problem sooner when it is easier and less expensive to repair
- Prevent the introduction of more errors
- Organizational learning from problems

Toyota Andon Cord



KEEP PUSHING QUALITY CLOSER TO THE SOURCE

Inefficient/Ineffective

Manual checks by a separate team that could be automated for the source team

Approvals by distant authorities who lack detailed knowledge

Detailed documentation that will quickly become obsolete

Sending large batches to special committees & waiting for processing or approval

Efficient & Effective

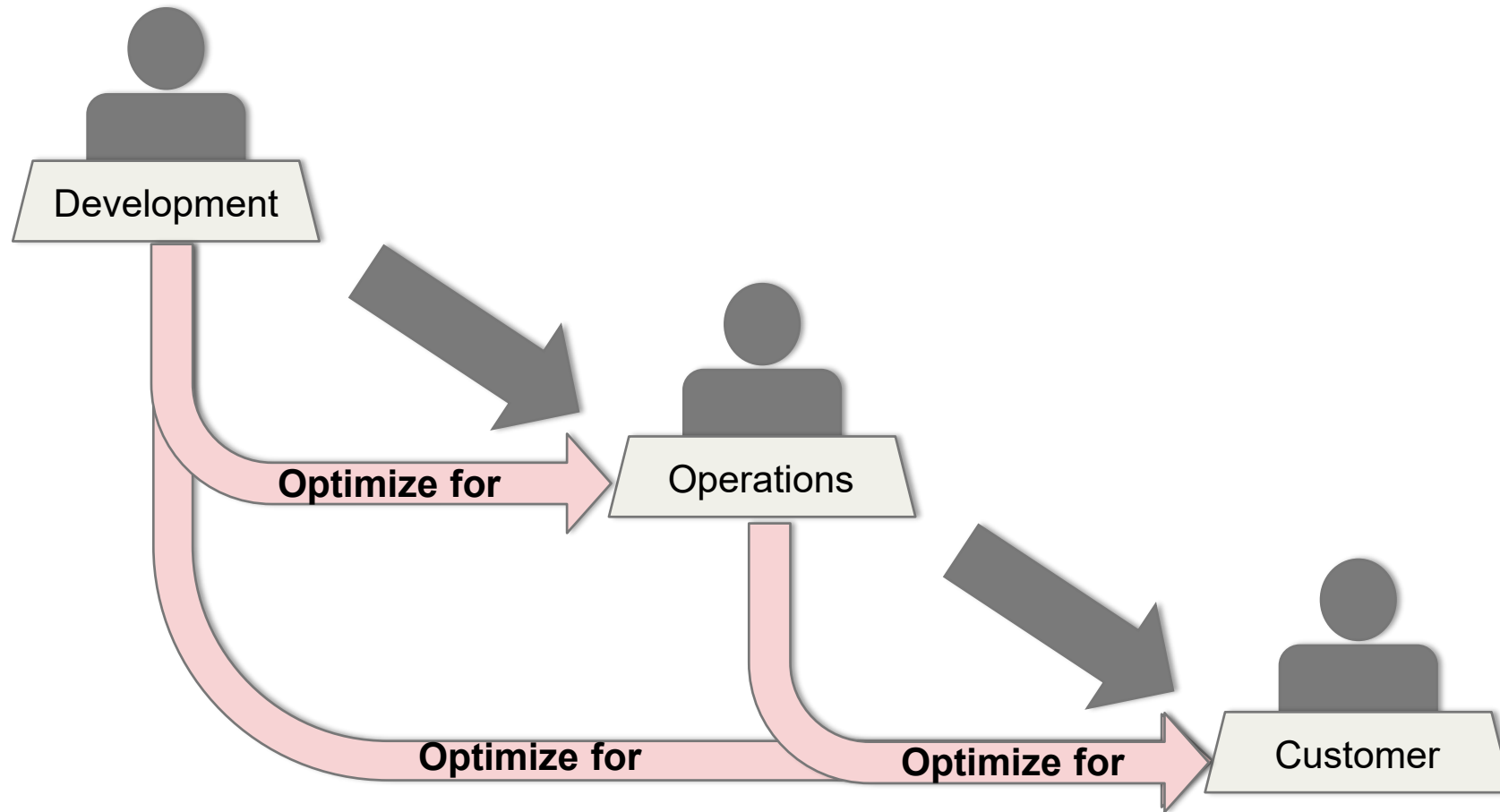
Enabling teams to find problems in their own work

Use Peer Reviews

Automate quality checks where possible

Make Quality each person's job

OPTIMIZE FOR DOWNSTREAM WORK CENTERS



THE THIRD WAY: CONTINUAL LEARNING AND EXPERIMENTATION

Principles of Continual Learning and Experimentation

- **Enable Organizational Learning and a Safety Culture**
- **Institutionalize the Improvement of Daily Work**
- **Transform Local Discoveries Into Global Improvements**
- **Inject Resilience Patterns Into Our Daily Work**
- **Leaders Reinforce a Learning Culture**

BUILDING A CULTURE OF LEARNING

**Failure is not a cause for blame,
it is a vehicle for change,
learning, and improvement.**

**If you don't fail sometimes...
You're not innovating!**

How do we get from:



To:



INSTITUTIONALIZE THE IMPROVEMENT OF DAILY WORK

The choice (processes never remain the same):

Degradation

Entropy

Workarounds

Growing Technical Debt

Trying to avoid disaster

Improvement

Reserve time every day:

- Pay down Technical Debt
- Fix defects
- Refactor problematic things

Kaizen Blitzes

- Attack bigger problems

INJECT RESILIENCE PATTERNS INTO OUR DAILY WORK

Normal daily work

(not in response to failure)

Reduce Lead Times

Increase test coverage

Improve productivity

Refactor for resilience

- Systems & Applications
- Processes & Methods

Special Exercises

“Game Day”

Rehearse large scale failures

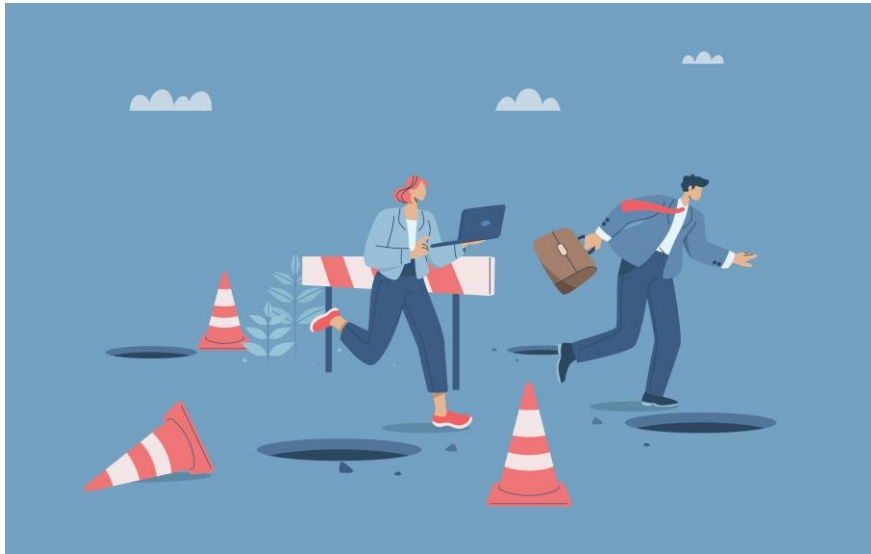
**Inject faults
(e.g. Chaos Monkey)**

CHECKLIST FOR DEVOPS SUCCESS

Champion Cultural Change

Prioritize Incremental Wins

Remove Organizational Barriers



INCREMENTAL STRATEGY



QUESTION 1

What is the primary purpose of the 'First Way' of DevOps?

- A. Continuous learning and improvement**
- B. Efficient flow of work from development to production**
- C. Automated decision-making**
- D. Eliminating feedback loops**

QUESTION 1

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QUESTION 2

Why is reducing batch sizes important in DevOps practices?

- A. It reduces collaboration**
- B. It leads to faster feedback and issue detection**
- C. It increases lead times**
- D. It decreases efficiency**

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QUESTION 3

What is a key reason for limiting work in progress (WIP) in DevOps?

- A. To encourage task switching**
- B. To increase the time needed for each task**
- C. To improve focus and reduce lead times**
- D. To increase bottlenecks**

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QUESTION 4

Why is visibility of work important in DevOps practices?

- A. To reduce transparency**
- B. To enable better collaboration and identify bottlenecks**
- C. To discourage team communication**
- D. To increase workload unpredictability**

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Why is visibility of work important in DevOps practices?

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QUESTION 5

What is the purpose of reducing handoffs in DevOps workflows?

- A. To increase delays and errors**
- B. To improve task ownership and reduce lead times**
- C. To encourage waiting for approvals**
- D. To create more dependencies**

QUESTION 5

What is the purpose of reducing handoffs in DevOps workflows?

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QUESTION 6

What is the 'Second Way' of DevOps primarily focused on?

A. Continuous feedback loops

B. Efficient automation

C. Large batch processing

D. Siloed team structures

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QUESTION 7

What does the 'Third Way' of DevOps emphasize?

A. Static processes

B. Continuous learning and experimentation

C. Avoiding failure at all costs

D. Limiting innovation

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QUESTION 8

What is the role of leadership in creating a successful DevOps culture?

- A. Promoting isolated teams**
- B. Championing collaboration and transparency**
- C. Avoiding organizational change**
- D. Increasing bureaucratic approvals**

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AGENDA FOR SESSION 2

Completing our introduction to DevOps with practical insights.

- **Recap of Session 1**
- **DevOps History and Evolution**
- **Major DevOps Tools and Platforms**
- **Infrastructure as Code Overview**
- **Collaboration and Communication**
- **Metrics for Success**
- **Group Exercise: DevOps Expectations**

RECAP OF SESSION 1



Key Takeaways from Last Session

DevOps Definition and Culture

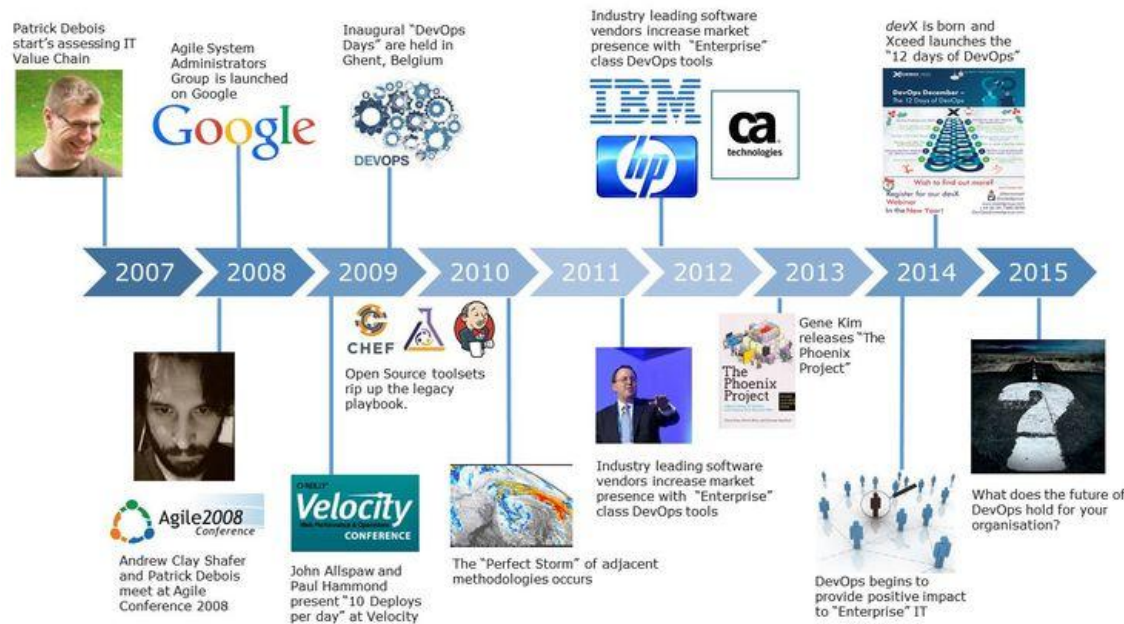
Three Ways: Flow, Feedback, Learning

Initial Benefits Explored



❖ DEVOPS HISTORY AND EVOLUTION

DevOps Timeline



- Origins of DevOps
- Evolution Over Time
- Industry Adoption Trends

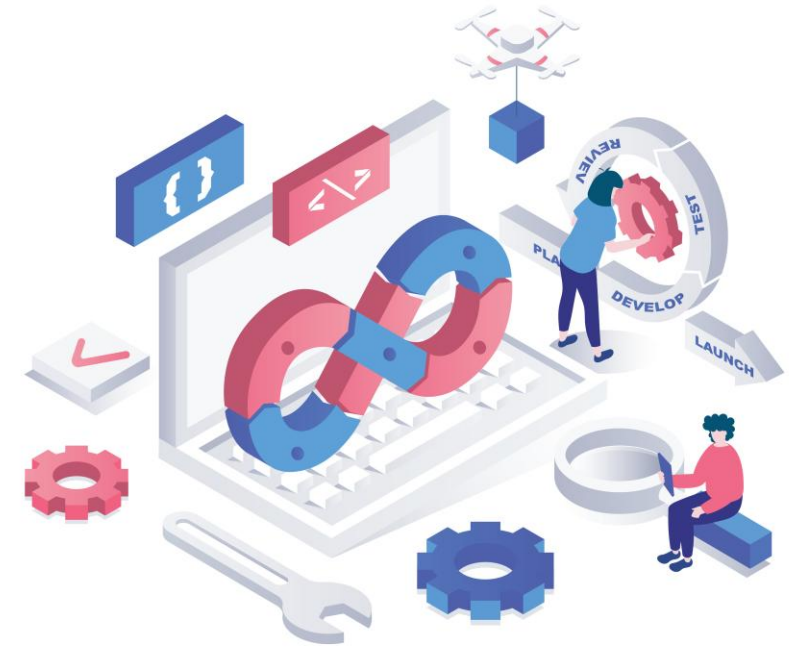
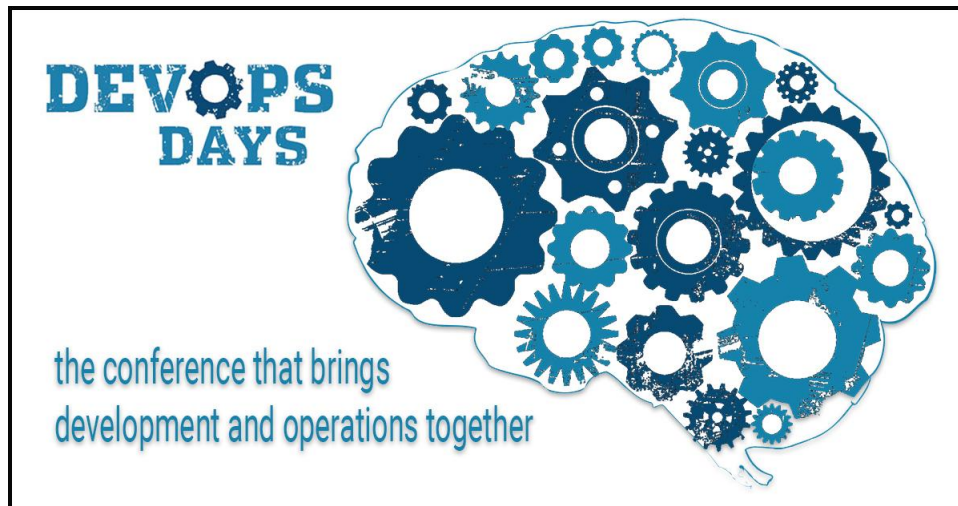
ORIGINS OF DEVOPS

The Birth of a Movement

Agile and IT Pain Points

Patrick Debois and DevOpsDays

Early Influencers



EVOLUTION OVER TIME

Growth of DevOps Practices

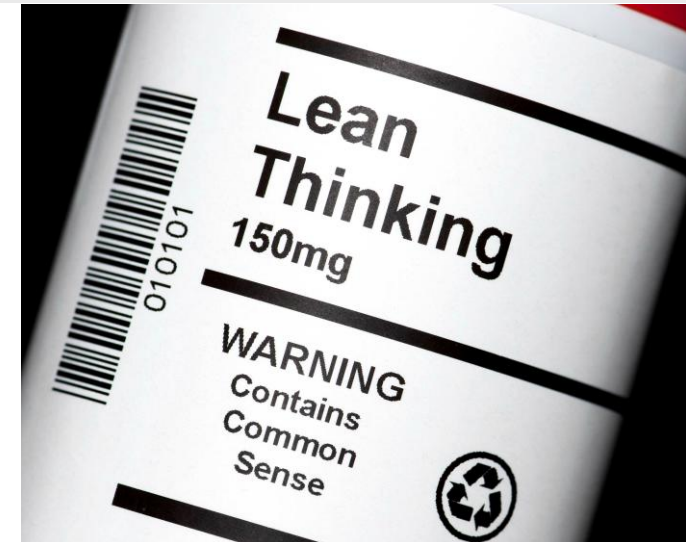
Integration with Lean and Agile

Rise of Automation

Cultural Shift Emphasis



Automation



INDUSTRY ADOPTION TRENDS

DevOps Goes Mainstream

Tech Giants Lead the Way

Cross-Industry Spread

Current State

NETFLIX



❖ MAJOR DEVOPS TOOLS AND PLATFORMS



Automation Tools
Collaboration Platforms
Monitoring Solutions



AUTOMATION TOOLS

Streamlining Processes

Jenkins for CI/CD/CD

Docker for Containers

Ansible for IaC



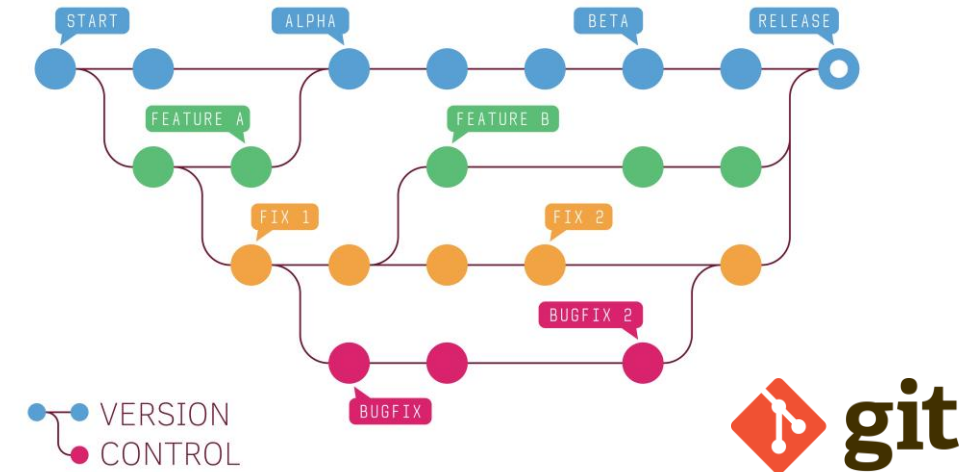
COLLABORATION PLATFORMS

Enhancing Teamwork

Slack for Communication

Git for Version Control

Jira for Task Management



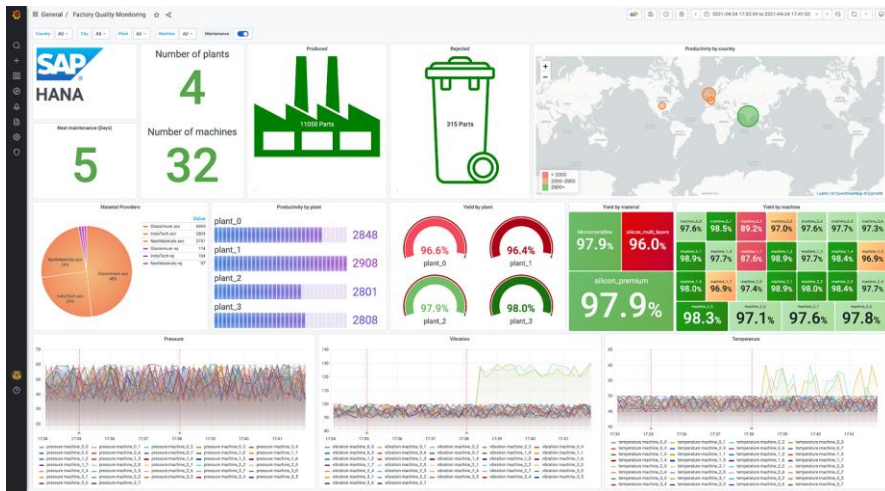
MONITORING SOLUTIONS

Ensuring Visibility

Prometheus for Metrics

Nagios for Alerts

Grafana for Dashboards



❖ INFRASTRUCTURE AS CODE OVERVIEW



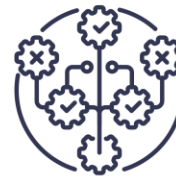
- What is IaC?
- Why It Matters
- Basic Examples



INFRASTRUCTURE AS CODE



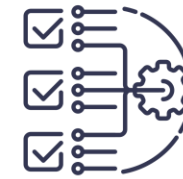
STREAMLINED
PROCESSES



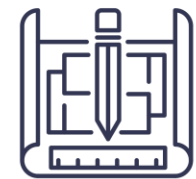
CHANGE
MANAGEMENT



SCALABILITY



UNIFORMITY



CONFIGURATION

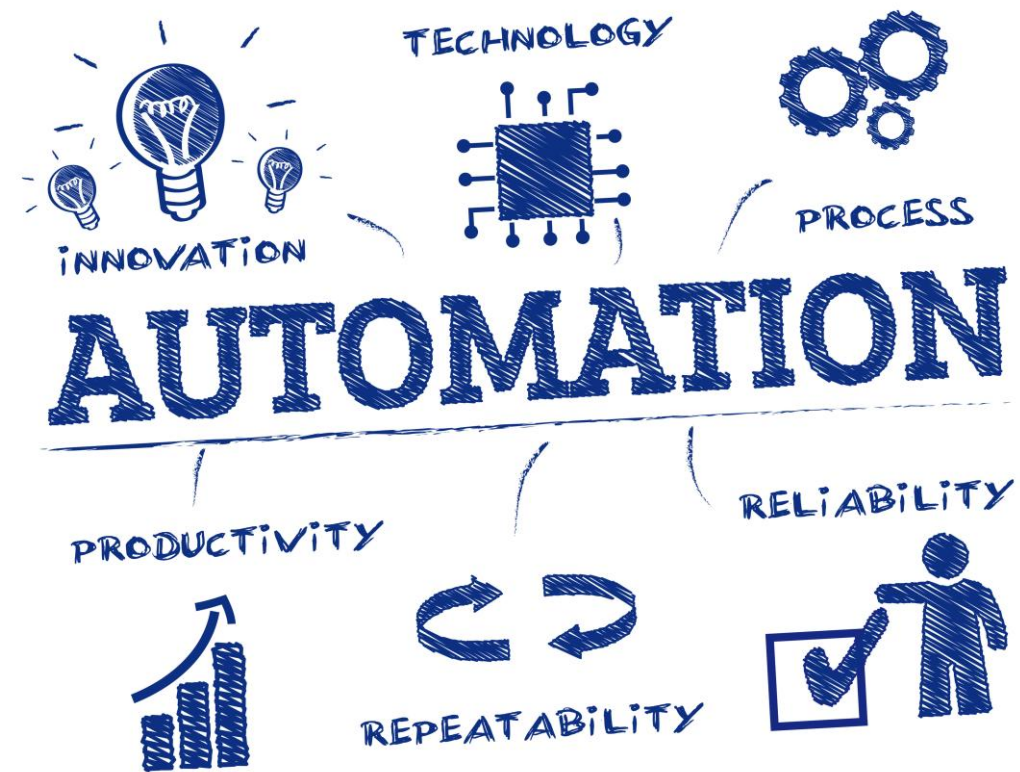
WHAT IS IAC?

Defining Infrastructure as Code

Code-Driven Infrastructure

Declarative Approach

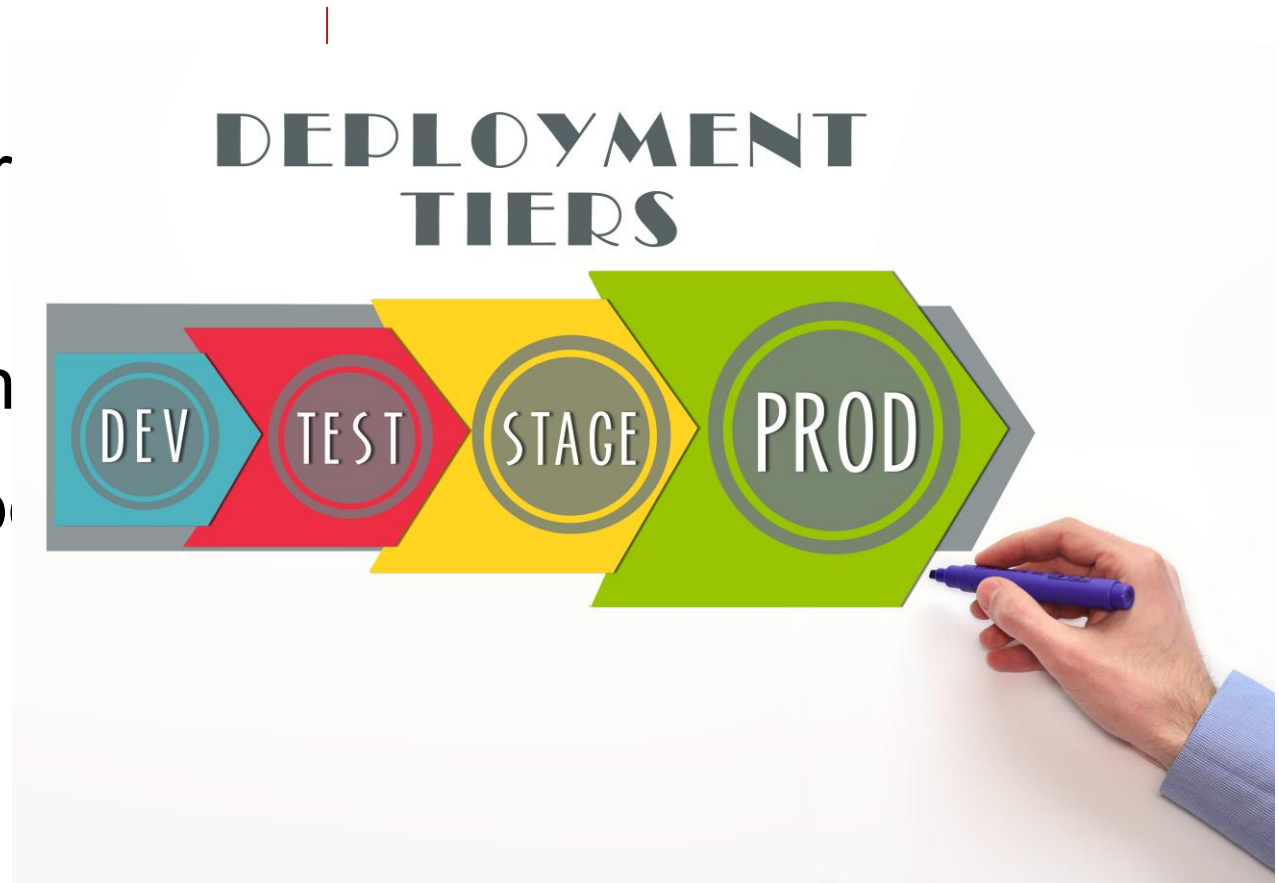
Automation Focus



WHY IT MATTERS

Benefits of IaC

- Consistency Across Environments
- Faster Provisioning
- Scalability Support



BASIC EXAMPLES



ANSIBLE

laC in Action

- Terraform Example
- Ansible Playbook
- CloudFormation Snippet



AWS CloudFormation

BASIC TERRAFORM CONFIG

```
resource "aws_instance" "Web" {  
  ami          = "ami-07013dd48140efd73" # Replace with your desired AMI ID  
  instance_type = "t3.micro"  
  tags = {  
    Name = "Ubuntu Server"  
  }  
  # subnet_id    = "vpc-0bf039e4e823f5223" # Replace with your subnet ID  
  key_name       = "AWS_Ubuntu_RSA"  
  # security_groups = ["Everybody All In"] # Ensure the security group exists with  
  this name  
}
```



BASIC ANSIBLE PLAYBOOK

- hosts: all

become: yes

tasks:

- name: Task - 1 Update APT package manager repositories cache

become: true

apt:

update_cache: yes

- name: Task -2 Install Java using Ansible

become: yes

apt:

name: "{{ packages }}"

state: present

vars:

packages:

- openjdk-11-jdk



❖ COLLABORATION AND COMMUNICATION



- Breaking Down Silos
- Tools for Collaboration
- Cultural Shifts



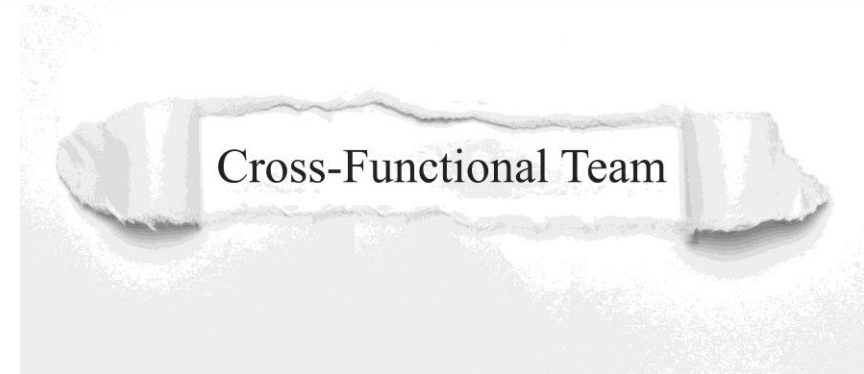
BREAKING DOWN SILOS

Fostering Cross-Functional Teams

Shared Goals

Cross-Team Metrics

Joint Decision-Making



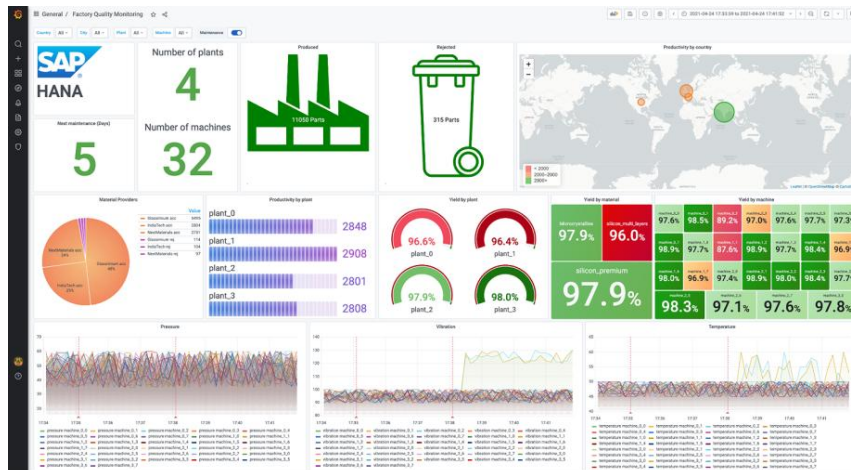
TOOLS FOR COLLABORATION

Enhancing Team Interaction

Slack, Teams

Kanban Boards

Shared Dashboards



CULTURAL SHIFTS

Promoting DevOps Culture

Continuous Learning

Blameless Culture

Innovation Encouragement



❖ **METRICS FOR SUCCESS**

- **Key Performance Indicators**
- **Aligning Metrics with Business**
- **Measuring Impact**



KPI

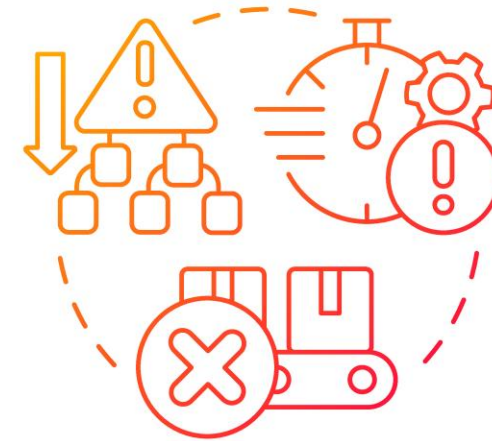
KEY PERFORMANCE INDICATORS

Understanding DevOps Metrics

Lead Time

Deployment Frequency

Change Failure Rate



**POOR BUILD QUALITY
& RELEASE SPEED**

ALIGNING METRICS WITH BUSINESS

Tying DevOps to Business Goals

Efficiency

Customer Satisfaction

Risk Reduction



MEASURING IMPACT

Quantifying DevOps Benefits

Before and After Comparisons

Benchmarking

Continuous Improvement



BEFORE

AFTER

**CONTINUOUS
IMPROVEMENT**



GROUP EXERCISE: DEVOPS EXPECTATIONS

Form Groups

Discuss Expectations

Present Insights



QUIZ QUESTION 1

What sparked the DevOps movement?

- A. New hardware advances
- B. Agile adoption challenges
- C. Regulatory changes
- D. Cloud cost reductions

QUIZ ANSWER 1

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QUIZ QUESTION 2

Which tool is used for containerization?

- A. Jenkins
- B. Docker
- C. Slack
- D. Prometheus

QUIZ ANSWER 2

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QUIZ QUESTION 3

What does IaC improve?

- A. Manual configuration time
- B. Consistency across environments
- C. Team silos
- D. Deployment errors

QUIZ ANSWER 3

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QUIZ QUESTION 4

What does visibility of work enable?

- A. Reduced collaboration
- B. Better bottleneck identification
- C. Increased delays
- D. Less accountability

QUIZ ANSWER 4

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QUIZ QUESTION 5

What is a DevOps cultural tenet?

- A. Siloed teams
- B. Continuous learning
- C. Blame assignment
- D. Static processes

QUIZ ANSWER 5

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QUIZ QUESTION 6

What does deployment frequency indicate?

- A. System downtime
- B. Release agility
- C. Error rates
- D. Manual effort

QUIZ ANSWER 6

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QUIZ QUESTION 7

Why reduce handoffs in DevOps?

- A. Increase delays
- B. Improve task ownership
- C. Add dependencies
- D. Slow delivery

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